BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 062 · Lesson 062 · Thinking Procedurally: Ordered Subtasks

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a precondition?

Q2. What is decomposition?

Q3. What is a dependency in a process?

Q4. 019

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

procedural thinking — Planning ordered steps.

subprocedure — Smaller named piece of a solution.

dependency — A stage depends on another.

atomicity — All required updates succeed or none do.

New learning

Identify — Identify the separate components of a problem: validation, stock check, updating records and confirming a result.

Model — Determine necessary order from dependencies. Do not send success or deduct stock before validation.

Evaluate — Decompose large operations into subprocedures, each with clear responsibility; update-related steps may need a single transaction to maintain consistency.

Worked example

Put steps in order to process an exam result securely: check identity, find result, return data.

1. Explain the decision — If lookup occurs before checking authorisation, data can be exposed through insecure responses.

We Do · Practise with guidance

A checkout program has steps deduct stock, check payment, pack goods. Choose order and justify.

```
Validate order → authorise payment → atomically confirm order and stock → pack
   or dispatch.
```

Explain the decision

Payment and stock operations need failure handling; no dispatch before a confirmed order.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Order tasks: issue library loan, check stock, validate pupil, generate receipt. Explain first and last. [6]

Question 2

Write three helper procedure names for the library process, each with one responsibility. [6]

Question 3

A system sends confirmation before saving a booking, then crashes. Explain fault and propose safer steps. [6]

Exit Ticket · Answer before checking

A confirmation should be sent before a transaction succeeds, to make an application feel faster.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Rule that must hold before an operation.

Q2. Breaking a problem into manageable subtasks.

Q3. A stage requiring another stage's completed result.

Q4. Stack Push 3,Push 5,Pop then Push8. Give top and popped value.

You Do pages · Independent questions

Q1. Validate pupil [2], check stock [1], issue record [1], receipt [1], last only after success [1].

Q2. ValidateBorrower() [2], CheckAvailability() [2], RecordLoan() [2] with clear purpose in each description.

Q3. False confirmation [2], validate [1], transactionally save/commit [2], send confirmation only after success [1].

Exit Ticket · Model answer

False. It can falsely confirm a failed transaction.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: